

B.Tech III Year II Semester (R20) Regular Examinations August 2023

DIGITAL SIGNAL PROCESSING

(Electrical & Electronics Engineering)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
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| (a) Draw the block diagram of digital signal processing system. | 2M |
| (b) What is the condition for z-transform to exist? | 2M |
| (c) What is decimation in frequency FFT? | 2M |
| (d) State any two properties of DFS. | 2M |
| (e) What are the different design techniques available for IIR filters? | 2M |
| (f) What is the main advantage of direct form II realization when compared to direct form I realization? | 2M |
| (g) What are the effects of windowing? | 2M |
| (h) What are the differences between IIR and FIR filters? | 2M |
| (i) What is multirate DSP? | 2M |
| (j) What is quantization error? | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 (a) Explain the classification of discrete-time signals. 5M
 (b) Determine the impulse response $h(n)$ for the system described by the second order difference equation $y(n) - 2y(n-1) = x(n) + x(n-1)$. 5M
- OR**
- 3 Explain in detail about the classification of Discrete Time Systems. 10M
- 4 Given $X(k) = \{36, -4 + j9.656, -4 + j4, -4 + j1.656, -4, -4 - j1.656, -4 - j4, -4 - j9.656\}$, find $x(n)$. 10M
- OR**
- 5 (a) State and prove any two properties of DFT. 4M
 (b) Given $x(n) = \{1, 2, 2, 3, 3, 2, 2, 1\}$. Find $X(k)$ using DIT FFT algorithm. 6M
- 6 Realize the system given by difference equation;
 $y(n) = -0.1y(n-1) + 0.71 y(n-2) + 0.7 x(n) - 0.252 x(n-2)$ in Cascade form and Parallel form. 10M
- OR**
- 7 (a) Compare Butterworth and Chebyshev Filters. 4M
 (b) Discuss in brief about the basic structures of IIR filters. 6M
- 8 Explain in brief about the different window functions used in FIR filter design. 5M
- OR**
- 9 Design a filter with

$$H_d(e^{j\omega}) = \begin{cases} e^{-j3\omega}, & -\frac{\pi}{4} \leq \omega \leq \frac{\pi}{4} \\ 0, & \frac{\pi}{4} \leq |\omega| \leq \pi \end{cases}$$
 Using Hamming window with $N = 7$. 10M
- 10 Explain about Round-off effects in digital filters. 10M
- OR**
- 11 Explain the following:
 (a) Decimation by a factor D. 5M
 (b) Interpolation by a factor I. 5M
